

Name: _____ Date: _____

Period: _____

Parametrization Practice

1. The ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ is centered at the origin with $a = 3$ and $b = 5$.

(a) Write the parametric equations that trace this ellipse counterclockwise, $0 \leq t \leq 2\pi$.

$x(t) = \underline{\hspace{2cm}}$ $y(t) = \underline{\hspace{2cm}}$

(b) Find the coordinates of the four extreme points of the ellipse using your parametrization. Show the t -value for each.

Extreme point	t	Coordinates
Rightmost	<u> </u>	<u> </u>
Leftmost	<u> </u>	<u> </u>
Topmost	<u> </u>	<u> </u>
Bottommost	<u> </u>	<u> </u>

2. The ellipse $\frac{(x-3)^2}{16} + \frac{(y+1)^2}{4} = 1$.

(a) Identify h , k , a , b , then write the parametric equations. _____

(b) Use your parametrization to find all t -values where the ellipse crosses the x -axis (set $y(t) = 0$ and solve). Then state the coordinates of those points. _____

Parabola Parametrization

3. The parabola $y = 2(x-1)^2 - 3$. Let $x(t) = 1 + t$, so $y(t) = 2t^2 - 3$.

(a) Find the x -intercepts of the parabola: set $y(t) = 0$ and solve for t . Then find the corresponding x -values.

(b) What is the vertex of the parabola? Use $t = 0$ in the parametrization to confirm. _____

4. The circle $x^2 + y^2 = 49$.

(a) Write the parametric equations. $x(t) = \underline{\hspace{2cm}}$ $y(t) = \underline{\hspace{2cm}}$

(b) Find the coordinates when $t = \pi/3$. (Use $\cos(\pi/3) = 1/2$, $\sin(\pi/3) = \sqrt{3}/2$.) _____

AP-Style Practice

5. Which of the following parametrizations correctly traces the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ counterclockwise for $0 \leq t \leq 2\pi$?

(A) $x(t) = 9 \cos t$, $y(t) = 25 \sin t$

(B) $x(t) = 5 \cos t$, $y(t) = 3 \sin t$

(C) $x(t) = 25 \cos t$, $y(t) = 9 \sin t$

(D) $x(t) = 3 \cos t$, $y(t) = 5 \sin t$